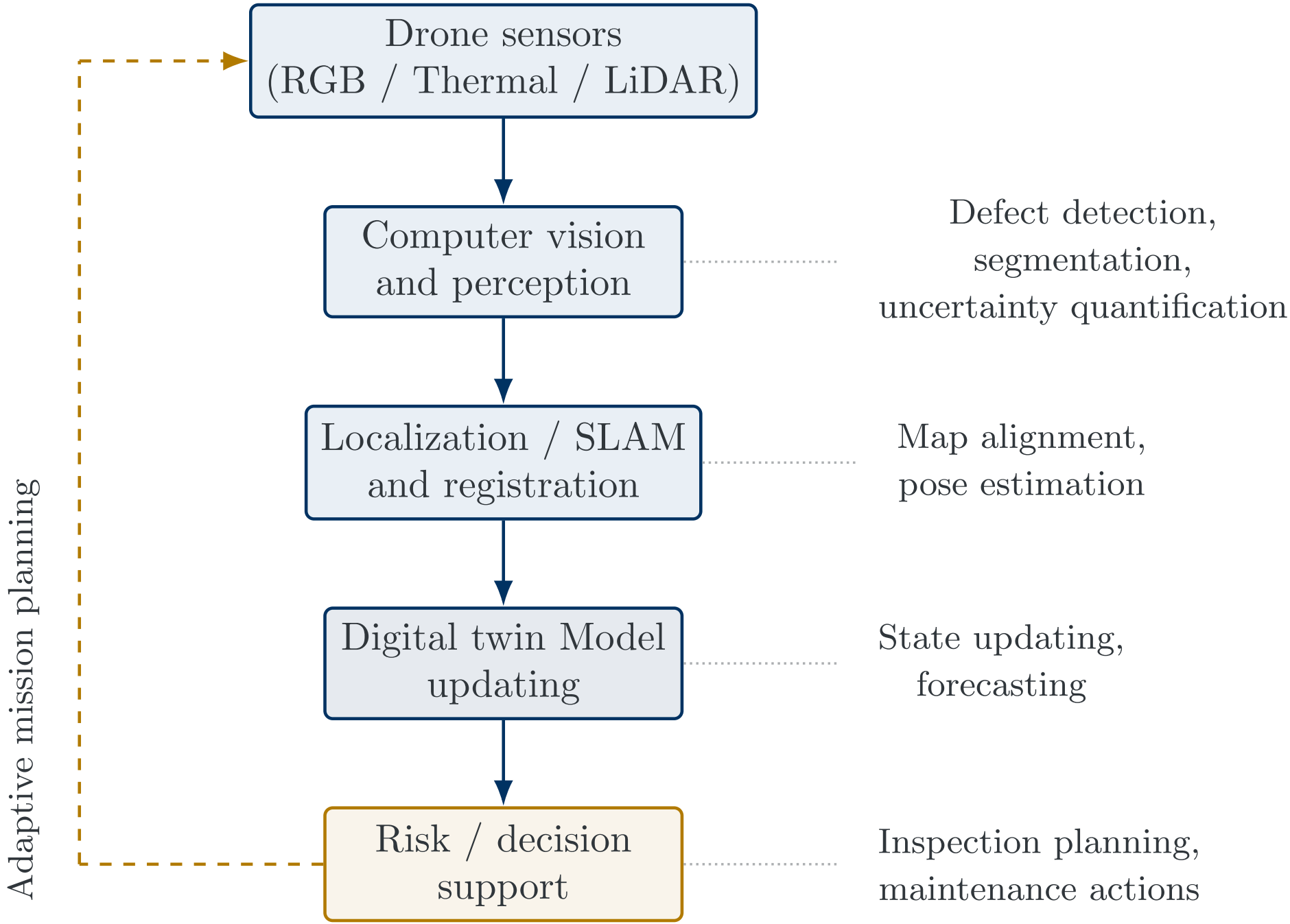
Drone sensors
(RGB / Thermal / LiDAR)
Computer vision
and perception
Defect detection,
segmentation,
uncertainty quantification
Localization / SLAM
and registration
Map alignment,
pose estimation
Digital twin Model
updating
State updating,
forecasting
Risk / decision
support
Inspection planning,
maintenance actions
Adaptive mission planning